EV Highway Energy Consumption Prediction Using Road and Traffic Context

1st Veerisetty Raaga Praneetha
Dept of CSE
Koneru Lakshmaiah Education Foundation
Guntur, India
2300031022@kluniversity.in

1st B.Jyothi
Dept of EEE
Koneru Lakshmaiah Education Foundation
Guntur, India
jyothieee@kluniversity.in
0000-0002-8019-1198

2nd Bhavana Pabbuleti
Dept of EEE
Koneru Lakshmaiah Education Foundation
Guntur, India
bhavana8099@gmail.com
0000-0003-4811-7665

2nd K.Sarada
Dept of EEE
Koneru Lakshmaiah Education Foundation
Guntur, India
saradak@kluniversity.in
0000-0002-3791-1014

1st Yadlapalli Aasritha Lakshmi
Dept of CSE
Koneru Lakshmaiah Education Foundation
Guntur, India
2300031056@kluniversity.in

1st Sankaramanchi Venkata Naga Sravya
Dept of CSE
Koneru Lakshmaiah Education Foundation
Guntur, India
2300031178@kluniversity.in

1st Devagiri Lalithya
Dept of CSE
Koneru Lakshmaiah Education Foundation
Guntur, India
2300031751@kluniversity.in

***Abstract*— Electric vehicles (EVs) have become an essential component in today's world of modern transportation due to their huge environmental and economic benefits. However, predicting the energy consumption of EVs, especially on highways, where the charging stations are irregularly available, remains a challenge because of constantly changing roads and traffic conditions from place to place and time to time. This study proposes a context-aware energy prediction framework for estimation of energy consumption of EVs by considering various influencing dynamic parameters such as road gradient, speed of the vehicle, traffic flow, and driver's behavior along with environmental conditions. The system uses a physics-based prediction model that combines real-time sensor data on aerodynamic drag, rolling resistance, and SOC-based battery efficiency. The proposed model gives more accurate predictions than older static estimation methods, which didn't consider changing parameters to figure out how much energy was used. This is because it uses road gradients and live traffic context in the energy estimation pipeline. This eco-friendly and energy-saving method makes it easier to plan routes, manage batteries, and find charging stations for sustainable electric mobility. It also eases drivers' range anxiety [1].**

***Keywords:* Electric Vehicles, Machine Learning, Energy Consumption Prediction, Road Context, Traffic Analysis, Smart Transportation.**

I. Introduction

A. Motivation

EVs hold a significant position in the transition toward eco-friendly transportation. Since these gas-powered engines are phased out from the vehicle system, EVs manage to ensure no burning of fossil fuel, which when done causes less release of harmful exhaust gases as there is no engine. Instead, with such a shift in dependency level, they run on battery power only and therefore add zero emission to the atmosphere like traditional vehicles who depend on crude oils that nowadays barely exist. In India, the rising costs of fuels have made this transition. Moreover, a main issue for EV users is estimating energy usage correctly which is range anxiety- the uncertainty about whether the energy will last until the next charging station especially when it comes to the highway where driving conditions can vary from moment to moment as the road slope or traffic, together with different driving behaviors, weather, traffic, and acceleration patterns, change. Traditional range estimate systems take average speed and battery capacity into account but leave out dynamic factors like weather effects on vehicular performance, road gradient, driver's driving patterns using the vehicle, and movement irrespective of driving patterns to derive a better, more

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accurate estimate or road congestion levels. This frequently results in false predictions and induces "range anxiety" for the drivers. Thus, integrating these specifications through the road and traffic context into energy consumption prediction is essential for enhancing EV efficiency via features like estimating range and planning the vehicle's charging infrastructure.

B. Importance of Energy Consumption Prediction:

The prediction of energy consumption accurately will help the EV drivers to plan the trips with confidence and support to plan the most efficient charging stops. By predicting accurately, the charging station operators predict the demand which helps in avoiding overcrowded stations, long waiting times, and underused stations. For city planners, accurate predictions will also help in designing the broader charging infrastructure by getting the idea of where to build the charging stations, how many charging stations are needed, and how powerful they should be. Below is the synthetic EV trip data generation for which will generate the realistic ranges and output of energy estimation which is done by considering all the factors like trip distance, average speed, temperature, elevation gain, and trip duration.

```python
import pandas as pd
import numpy as np

np.random.seed(42)
n_samples = 500

# Generate features within realistic ranges
trip_distance_km = np.random.uniform(1, 200, n_samples)
average_speed_kmh = np.random.uniform(20, 120, n_samples)
outside_temp_c = np.random.uniform(-10, 40, n_samples)
elevation_gain_m = np.random.uniform(0, 1500, n_samples)

# Trip duration is distance / speed (hours) converted to minutes + some noise
trip_duration_min = (trip_distance_km / average_speed_kmh) * 60
trip_duration_min += np.random.normal(0, 5, n_samples)  # some variation

# Calculate synthetic energy consumption (kWh)
# Base consumption proportional to distance
base_energy = trip_distance_km * 0.15  # 0.15 kWh per km as base

# Increase consumption with elevation gain
elevation_factor = elevation_gain_m * 0.005

# Speed penalty: consumption rises at very high speed (> 80 km/h)
speed_penalty = np.where(average_speed_kmh > 80,
                         (average_speed_kmh - 80) * 0.2,
                         0)

# Temperature effect: increased consumption if too cold or hot
temp_penalty = np.where(outside_temp_c < 5,
                        (5 - outside_temp_c) * 0.1,
                        np.where(outside_temp_c > 30,
                                 (outside_temp_c - 30) * 0.1,
                                 0))

# Add some random noise
noise = np.random.normal(0, 2, n_samples)
```

Fig. 1.1 Python Code for Synthetic EV Trip Data Generation and Sample Output Table

•••	trip_distance_km	average_speed_kmh	outside_temp_c	elevation_gain_m
0	75.533484	89.816171	-0.743354	778.622678
1	190.192147	73.609637	17.095047	718.772816
2	146.666794	50.952762	33.647292	38.463099
3	120.133038	101.379502	26.611244	511.871741
4	32.047709	88.473117	30.328057	570.293428

	trip_duration_min	energy_consumption_kwh
0	46.068807	16.444473
1	150.893246	31.515234
2	171.576746	19.865322
3	72.935838	23.216699
4	26.301786	8.433611

Fig:1.2 Output of energy estimation through the parameters- trip distance, average speed, temperature, elevation gain and trip duration.

The above figure shows sample readings of an electric vehicle (EV) trip dataset containing around six variables: trip distance (km), average speed (km/h), atmosphere temperature (°C), elevation gain (m), trip duration (min), and energy consumption (in kWh).

Each row represents an individual trip with the corresponding measurements for these parameters. These parameters help in predicting energy consumption in the following way:

Trip distance – It helps the driver to estimate how much energy is required to reach the destination and plan the charging spots correspondingly [4].

Average Speed – Usually at high speeds, the force required to move the car forward requires more energy due to an increase in aerodynamic drag.

Temperature – This is considered as the most critical parameter that influences the EV's energy. Less environmental temperature, i.e., cold weather particularly increases energy consumption. It highly reduces the battery's capacity (chemical ability) to store and deliver energy to the parts of the EV. Additionally, it must maintain both the temperature of the electric vehicle and battery's state too.

Elevation Gain – To move up on an uphill on the highway, the vehicle needs to oppose the gravitational force that keeps dragging the vehicle downwards. Hence, it requires more energy, and it increases the load on the motor. Whereas, during the downhill, the vehicle uses regenerative braking to recharge the battery and reduce consumption of energy.

Trip duration – Generally, extra features like air conditioning and wipers along with infotainment systems also require energy from the battery. Basically, the trip with long distances and with low speed consumes more energy than the trip with the same low speed to cover a short distance [5], as it must provide for these auxiliary loads for more time.

C. Role of Road and Traffic context in EV Energy Usage

The energy consumption of an area does not solely depend on distance travel. Temperature, weather conditions, road gradient, acceleration patterns, traffic dynamics, including congestion, frequent stops, and varying speed limits, different driving behaviors, and other factors influence the energy drawn from the battery.

For example, driving in chilly weather can increase consumption by 20-30% as the vehicle uses an electric heater, which is more energy-intensive than AC. When driving uphill the consumption will increase as extra energy is needed to lift the vehicle upwards.

The existing prediction system only focuses on traditional methods like speed, vehicle mass, road slope and air resistance which result in inaccurate estimations. This approach below uses real-time sensor data by capturing the factors to improve prediction accuracy.

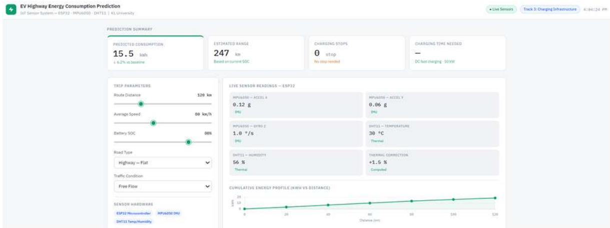

Fig.1.3.1: EV energy prediction Dashboard

II. IMPLEMENTATION

A. The following image represents a real-time EV Highway Visualization that is simulated by considering parameters like speed, acceleration and road gradient. It showcases a red-colored electric vehicle travels down a particular virtual highway, accompanied by a graph that forecasts energy consumption, which gradually decreases in time. The simulation takes real-world parameters into cosideration, including elevation gain, driving speed, and temperature to estimate energy usage. Moreover, it also displays the speed (km/h), acceleration (m/s^2) and the slope value on the top left. On the right side, a part of Python code is written and displayed [6]. Overall, this provides both visual and analytical insights of an electric vehicle performance on highways.

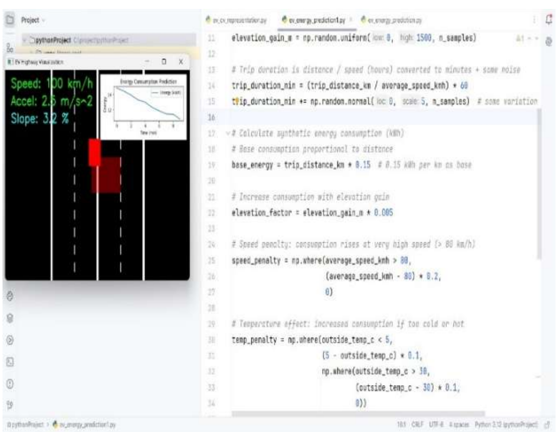

Fig:2.1 Dynamic EV Highway Simulation Showing Speed, Slope, and Energy Consumption Trends

B. The below images represent the real-time graph of energy consumption prediction using dynamic parameters-acceleration, traffic density, road-type, battery level. A live dataset is taken from Kaggle and random values of the considered parameters are generated through random() function, which finally outputs the energy consumption by creating a visual representation of consumption of the energy in an EV at different cases and scenarios.

When the graph line goes up, it shows that energy consumption is increased. This generally happens when the acceleration is raised; traffic density is high (commonly between 0.8-1.0), road type like- uphill and downhill, and low battery health conditions.

The graph line goes down when there is negative acceleration (known as regenerative braking), low traffic density (mostly ~0.1), and the road with less/no uphill and downhill.

This uncertainty in the graph happens every 0.8 seconds. The graph will be updated with new acceleration values and a random set of driving conditions from the dataset. This simulation model continuously predicts range, efficiency, and battery drain.

In this, a random forest model is created with one hundred decision trees:

n_estimators=100

Each tree learns continuous patterns between random values of acceleration, traffic density, road type, and battery level. This model uses Random feature selection and creates decision splits to predict energy values. The random forest is trained at:

model.fit(X_train, y_train)

The final output will be the average of all predictions from all the trees through:

predicted_energy = model.predict(sample_df)[0]

This simulation model is like the systems in BYD, Tesla.

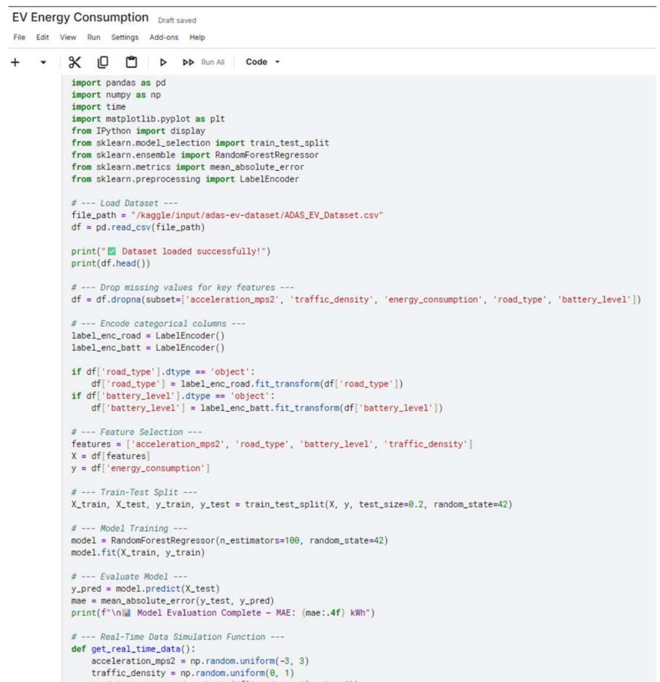

Fig. 2.2.1: Python code for real-time energy prediction graph

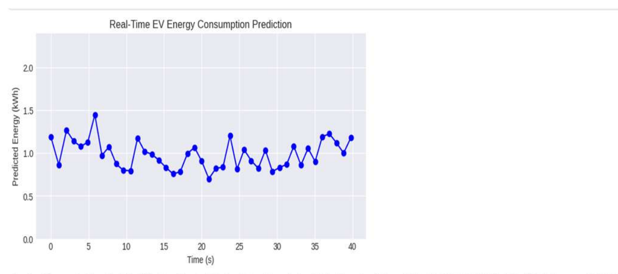

Fig:2.2.2 Real-time graph using dynamic parameters for energy prediction

C. Below (fig.2.3.1) dashboard showcases the prediction summary of the EV vehicle giving an overview of vehicle status like the acceleration values, temperature, and the trip details. The distance is provided, it calculates the speed, by continuously monitoring the battery SOC. By this, it gives the number of charging stops which the vehicle needs to halt at, to make sure the destination is reached without any charging issues, by reducing the anxiety among drivers. It uses live APIs to track the real-time road types and traffic conditions, and all together, it gives the information about the cumulative energy that is being consumed by the vehicle in a graph (KWH vs Distance).

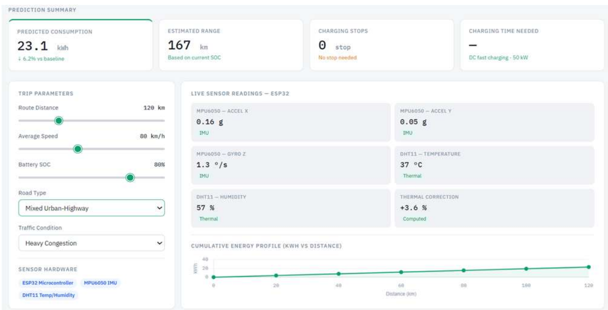

Fig. 2.3.1: Dashboard of Energy prediction summary

The next figure (fig.2.3.2) is a part of the dashboard showing the traffic density in a segment view. This view shows the following parameters:

- Traffic condition: Free flow
- Road type: Highway

The vehicle consumes energy in a consistent way since the car behavior doesn't change its state continuously.

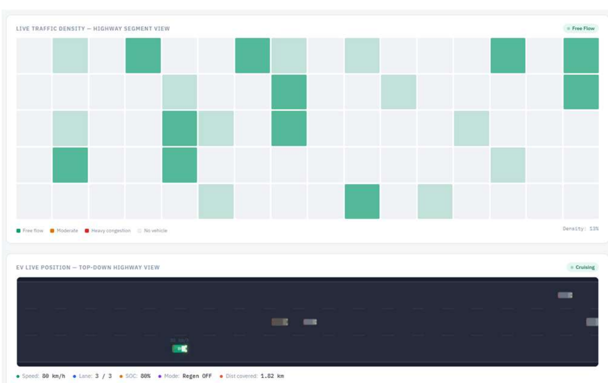

Fig.2.3.2: Semantic view of traffic density- Free flow

The following figure (fig.2.3.3) displays the traffic density with the parameters:

- Traffic condition: Heavy Congestion
- Road type: Mixed Urban Highway

The consumption of energy of the EV is quite abnormal since the vehicle must stop many times because of the heavy traffic congestion.

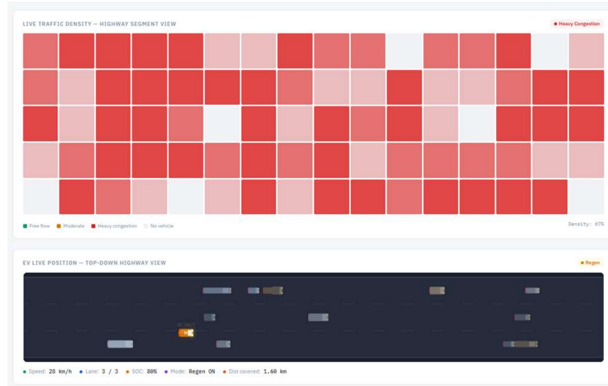

Fig.2.3.3: Semantic view of traffic density- Heavy congestion

The next figure shows the ML prediction grid, which uses the Physics-informed ML Ensemble model to show how much energy will be needed to get to the destination. It also gives us the efficiency of the energy that is being consumed. It provides 95% confidence intervals by using bootstrap uncertainty, by reporting feature contribution of aerodynamic drag loss, rolling resistance, thermal efficiency loss, traffic congestion penalty. This shows a typical style of Random Foorest Regression model.
It shows the energy breakdown for speed, road gradient, traffic and thermal congestion.
This dashboard also recommends charging stops in the way to the destination according to battery SOC.
For the record, it calculates the performance of other models with this proposed model to show the best accuracy among all.
To evaluate the proposed model, its performance was compared against three machine learning approaches: Linear Regression, Decision Tree, and Support Vector Regression (SVR). Evaluation metrics used were Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). The proposed model achieved the lowest RMSE (1.24 kWh) and MAE (0.98 kWh) with the highest R^2 value (0.94), demonstrating superior predictive accuracy over all baseline methods.

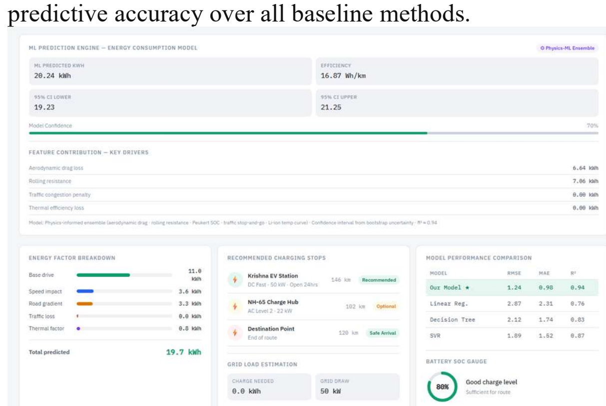

Fig.2.3.4: ML prediction model in dashboard

III. HARDWARE

MPU6050, DHT22 sensors data that are connected to the vehicle, are used to calculate real-time energy prediction, by considering the values of acceleration(x,y,z), gyro(x,y,z) and the temperature to predict energy. The data will be displayed through the serial monitor.

Driving behavior can also be estimated through MPU6050 (acceleration). Aggressive acceleration leads to energy reduction in the battery, while smooth acceleration reduces the load on the battery.

During aggressive driving, acceleration increases rapidly, which results in heavy loads on the motor. In this case, more current flows from the battery, leading to more internal heating. These results rise in the temperature of the battery. To reduce the battery temperature, the cooling systems are activated, which increases the overall consumption of energy in the EV. Acceleration determines how quickly the speed is changing [7]. If acceleration increases, the force that is required also increases, leading to an increase in battery energy consumption for the supply of current to the motor.

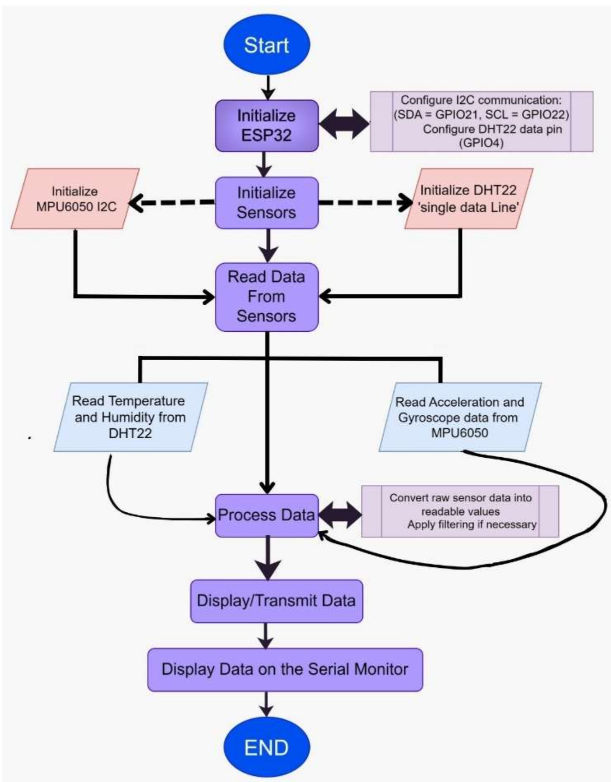

Fig 3.1: Energy Prediction through temperature, acceleration and gyro valuesusing MPU6050, DHT22 sensors

Following figure is a simulation-based execution, where MPU6050(to measure acceleration values), DHT22(for temperature) and GPS-neo-6M modules are used to obtain the value of speed of the EV, predict the energy that is being consumed at that point of time, based upon the values of the input data sensors.

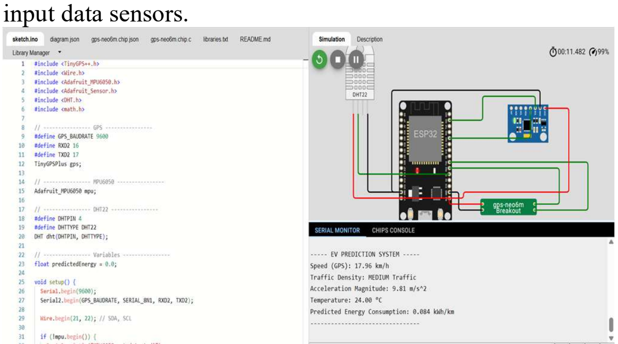

Fig 3.2: Simulation based energy consumption prediction

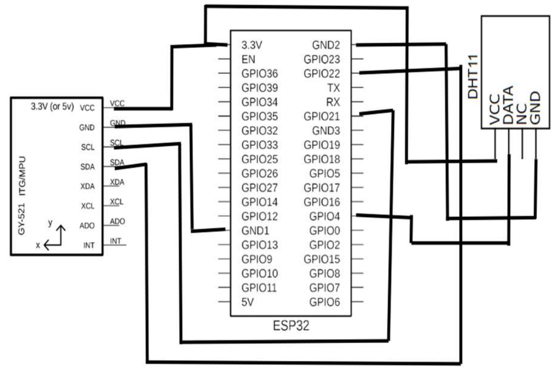

Fig 3.3: Block Diagram of connections (Components used: ESP32, MPU6050, DHT11)

For the actual hardware implementation, ARDUINO IDE sketch is used. It reads the temperature from the DHT11 sensor, simulates motion activity and gives the battery level. The real motion is detetcted from the MPU sensor.
For the battery-deplition logic:

- If the motion value is high, battery decreases by 10%

For the low charging in the battery, it gives a warning through the LED:

- If battery <=20%, LED blinks with delay(1000)
- Else LED stays off

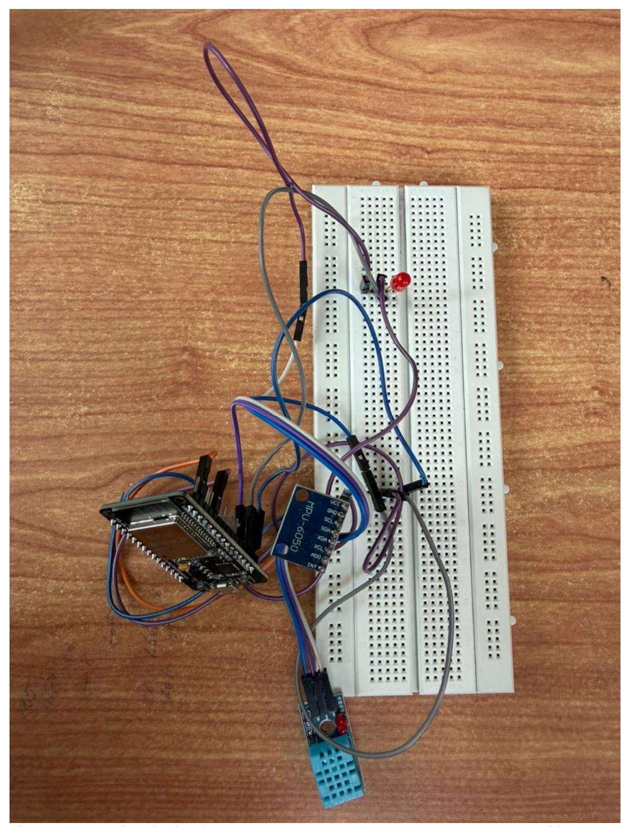

Fig 3.4: Integrating the hardware components- MPU6050, ESP32, DHT11 with LED

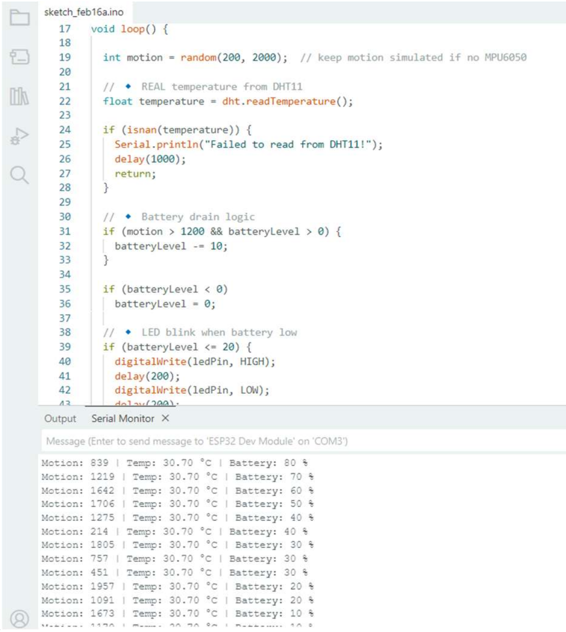

Fig 3.5: Implementation of the code

IV. METHODOLOGY

The proposed system architecture consists of - data acquisition, feature extraction, model training, and prediction, evaluation modules.

- Data Acquisition: Data is collected from ESP32 microcontroller.It reads the data from the two sensors-MPU6050, which gives the motion data and DHT11, which gives the temperature and humidity data
- Feature Extraction: Key features include road gradient, distance, vehicle mass, traffic density, and average velocity along with dynamic parameters.
- Model Training: Machine learning algorithms [9] such as Linear Regression, Random Forest regression and LSTM model networks are used to train based on historical data of the EV energy usage patterns.
- Prediction: The proposed model predicts the energy patterns by considering five factors which are base drive, speed impact, road gradient, traffic loss and thermal factor.

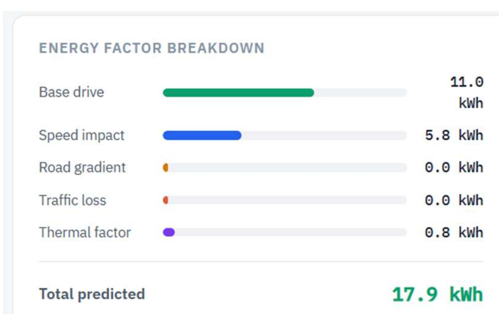

Fig 4.1 Energy Factor Breakdown

- Evaluation: The model's performance is based on the three baseline machine learning appraches: Linear Regression, Decision Tree and Support Vector Regression (SVR).The evaluation metrics which were used are Root Mean Square Error(RMSE), Mean Absolute Error(MAE) and the coefficient determination($R^{2)}$.

MODEL PERFORMANCE COMPARISON

MODEL	RMSE	MAE	R^2
Our Model ★	1.24	0.98	0.94
Linear Reg.	2.87	2.31	0.76
Decision Tree	2.12	1.74	0.83
SVR	1.89	1.52	0.87

Fig 4.2 Model Performance Comparision table-RMSE, MAE and R^2

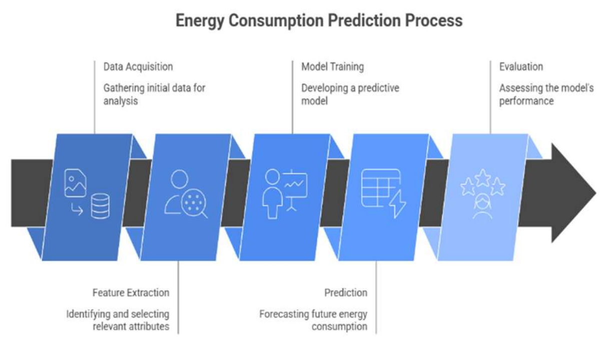

Fig:4.3 EV Highway Energy Consumption Prediction Process

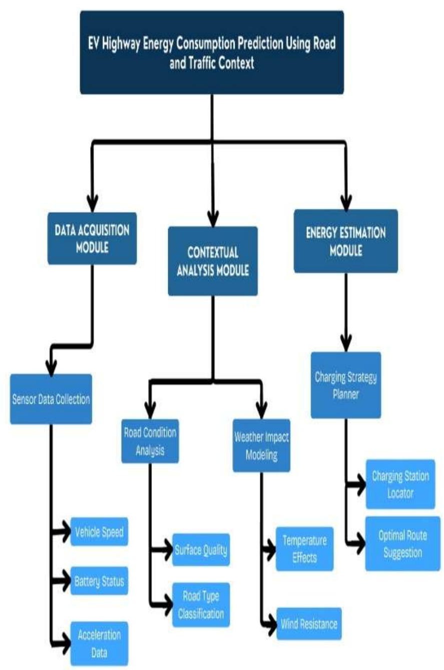

Fig:4.4 EV Highway Energy Consumption Prediction Using Road and Traffic Context Modelling and working flow

V. CONCLUSION

This project demonstrates a method for predicting EV highway energy consumption using a combination of parameters- road and traffic context. The proposed machine learning-based framework reduces estimation errors and enhances the robustness of energy management control systems in electric vehicles. In the implementation of hardware, it is demonstrated that the energy prediction is done by using Temperature and Acceleration sensors, and the LED signals when the battery reaches below 20%. Further, the future work will focus on deploying this system in real-time through IoT sensors installed in EV's along highways and integrating it into onboard vehicle systems for adaptive route and charging planning of the vehicle [8]. For more additional features, if charging slot is available in the designated location or not will be done for better planning of battery charging, without wasting time for the driver to reach the destination.

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